Daye Lee

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Education

Seoul National University — Seoul, Korea
M.S. in Data Science — *Mar 2022 - Aug 2025*

- Multi-modal Deep learning Theories and Application, Sep 23 - Dec 23
- Natural Language Processing, Conversational AI, Dialogue Systems, Sep 23 - Dec 23

University of Toronto — Toronto, Canada
IITP Visiting AI Researcher, Centre for Analytics and Artificial Intelligence Engineering — *Jan 2024 - Jun 2024*

- *Sponsored by Institute for Information & Communication Technology Planning & Evaluation (IITP), Korean Government Institute*
- Completed a 6-month research training in AI at CARTE; coursework + applied projects.
- Collaborated with Cyberworks Robotics on real-time RGB and RGB-D object detection and depth estimation with CPU optimizations for deployment.
- Mobile Robotics and Perception, Jan 24 - Apr 24

Seoul National University — Seoul, Korea
B.S in Applied Biology and Chemistry — *Feb 2022*

- GPA: 3.72/4.3

Cadillac High School — Michigan, U.S.
Study Abroad — *2013 - 2014*

Technical Skills

Languages (Advanced): Python, C/C++
Languages (Intermediate): SQL (PostgreSQL), MATLAB
Developer Tools: Git, Docker, K8s, VS Code, W&B, AWS, Jira, Confluence
Libraries & Frameworks: PyTorch, PyTorch3D, Accelerate, ROMP, VIBE, Linux, ROS
Software Tools: Blender, RViz

Professional Publications

Daye Lee, Minseok Kong, Jiho Park, Nikolaos Kourtzanidis et al. 2025. *Simulating Mobile Robot Vision: An Analysis of RGB-D versus RGB-Based Accuracy and CPU Optimization*. ICAIIC

Experience

Cyberworks Robotics, Autonomous self-driving company — Toronto, Canada
Project member of three students — *Jan 2024 – Jun 2024*

- Built a real-time object-detection + depth-perception pipeline for resource-constrained, CPU-only robots (e.g., smart wheelchairs and large cleaning robots).
- Processed live camera feeds in two modes—**RGB-only** (YOLOv8 + MobileNetV2-based monocular depth) and **RGB-D**—and fused outputs to produce 2D bounding boxes with per-object range estimates with Intel RealSense d435i RGB-D camera with a 640×480 pixels resolution.

- Visualized results in ROS/RViz with 3D bounding-box markers and bird's-eye-view (BEV) overlays for online depth sanity checks.
- Achieved real-time performance at 3 FPS on the 6 cores of Intel(R) i7-9750H CPU @ 2.60GHz and ROS noetic while reducing CPU usage by 7% (RGB-D) and 5% (RGB) via OpenVINO conversion + INT8 quantization (MobileNetV2).

MS THESIS

Controllable 3D Dance Motion Generation | *Diffusion Models, SMPL, Blender, ROMP*Sep 2024 – Aug 2025

- Proposed a **multi-condition controllable diffusion framework** for **3D dance motion generation**, integrating **music, dance genre, and sparse keyframe constraints**.
- Introduced **Chained Classifier-Free Guidance (CFG)** to disentangle and jointly control multiple conditions, achieving improvements over single-condition baselines in motion quality (**Jitteriness ↓ 1.27%**), genre expressivity (Top-1 +50%, F1 +15%), motion diversity ($Dist_k$ +1.4%), and **music–motion alignment (+9.4%**).
- Constructed **CrossGenreDanceSet** (128 min curated data) in **SMPL format** using ROMP, a 12-genre dataset with genre–music mismatched pairs, outperforming POPDG in beat alignment (0.3029 vs. 0.2499, **+21%**) and reducing motion jitter by **39%**.
- Designed **Gaussian-masked keyframe conditioning** for motion in-betweening, reducing transition jitter by **39%** and improving structural fidelity (MPJPE ↓ **41%**, $MPJPE_k$ ↓ **99.4%**).
- Proposed new evaluation metrics for motion in-betweening, including **structural fidelity and keyframe recall** (MPJPE, $MPJPE_k$, $MPJPE_g$, Recall@0.3).
- Developed a **Transformer-based dance genre classifier** trained on **GenreDanceSet** (201 min curated data), outperforming ground-truth annotations in genre consistency (Top-3 accuracy: 0.528 vs. 0.307, F1: 0.176 vs. 0.091).

PROJECTS

SmartCast Robotics | *ROS2 Jazzy, LeRobot(ACT, Diffusion, π0)* Mar 2026 – May 2026

- Developed a smart factory system for high-mix, low-volume sand casting production as part of a job training program at Addinedu.
- Currently developing a VLA pipeline with **LeRobot using ACT, Diffusion Policy, and SmolVLA**, while collecting real- robot demonstration data for project integration.
- Currently collecting real-robot demonstration data with a fixed Intel RealSense camera and kinesthetic teaching, then converted the dataset into LeRobot training format for **imitation learning** experiments.
- Integrated a PinkyPro AMR running on Raspberry Pi 4 and a Mycobot280 robotic arm running on Raspberry Pi 5 within a ROS 2-based automation workflow.
- Led a 10-member team as Scrum Master, managing sprint planning, task tracking, and technical documentation through Jira and Confluence in an Agile development cycle.
- Designed a relational database schema (ERD) for a Fleet Management System (FMS) and led database normalization up to 3NF.
- Improved code reliability by introducing Pydantic-based data validation, Pytest-driven test practices, and clear interface contracts across modules.
- Standardized GitHub branch strategy, PR review workflow, and CI conventions for a 10- person distributed team.

Emotion-Specialized Text-to-Video Retrieval | *Python, Pytorch* Jan 2024 – May 2024

- Developed a text-to-video retrieval system that specializes in extracting videos based on emotion-rich text queries
- Modified code to align with available distributed training library - accelerate, conducted model training experiments, and shared fine-tuned model checkpoints with the team
- Implemented evaluation metrics such as recall based on cosine similarities matrix to measure the similarity between the text query and the video embeddings in the dataset, retrieving the top-n videos in descending order of relevance
- Discussed the t-SNE result of embedding space and identified changes in the embeddings associated with similar emotions to be mapped closer together in the space
- Found a decline in overall quantitative performance when contrasting baseline with our methods, concluding that less training dataset led to the performance degradation
- Implemented demonstration code in Jupyter Notebook to showcase the project

Personalization of Music Conditioned Dance generation | *Python, Pytorch* Sep 2023 – Dec 2023

- Deployed few-shot learning methods such as Dreambooth, Inversion and etc, and fine-tuned the EDGE model in three approaches with three different few-shot dataset, a Diffusion-based baseline model where the model learns the correlation between music and SMPL pose data through cross-attention
- Generated few-shot dataset by crawling YouTube and processed the data with ROMP to extract SMPL information
- Resulted in decreasing FID_k by 81, FID_m by 2.432 and PFC by 0.98 and enhancing Beat Alignment by 0.074 and outperforming in human evaluation in all three approaches
- Demonstrated the qualitative motion results by rendering them using Blender

Dialog Inpainting for Legal Dialogue Systems | *Python* Sep 2023 – Dec 2023

- Utilized the dialog inpainting approach (Dai et al., 2022) as a data augmentation technique to construct a dataset suitable for developing a legal consultation chatbot.
- Identified three key characteristics of legal documents: hierarchical structure, inter-referencing structure, and similar-context structure.
- Created a question-answer dialogue dataset using the ChatGPT 3.5 model, generating 3,048, 3,408, and 474 conversations for the hierarchical, referencing, and similarity-based approaches, respectively.
- Evaluated the fine-tuned Llama2-7B chat model against a baseline model based on accuracy, informativeness, well-formedness, and overall quality across 10 conversations for the question classification (QC) versus question answering (QA) task and 15 conversations for legal domain-specific approaches, involving evaluation by all five team members.

Volunteer and Teaching Experience

Volunteer Translator, TED Translators Sep 2024 – Present

- Translate TED and TED-Ed talks from English to Korean (three published talks).

Instructor, Python for Algorithmic Problem Solving Sep 2025 -Nov 2025

- Teaching Python fundamentals and competitive programming algorithms
- Language of instruction: English with supplementary Korean support

Tutor, Introduction to Computing Mar 2025 – Aug 2025

- Tutored undergraduate students for two semesters in the course **Introduction to Computing: Foundations and Basics**.
- Recognized with the Outstanding Tutor Award for excellence in teaching support.

CERTIFICATES

PyTorch for Deep Learning with Python bootcamp Oct 2023

- NumPy, Pandas, Machine Learning, Model Evaluation, Tensors, Neural Network, ANN, CNN, RNN